

Enhancing Electricity Demand Forecasting in NSW Under Extreme Temperature Conditions

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Abstract

This report focuses on the robustness of electricity demand forecasting models during extremely hot days. Analysis was conducted on half-hourly electricity demand and temperature data for New South Wales from January 2010 to March 2021. Exploratory data analysis of current forecasts shows that the current forecast model performs worse on hot days compared to normal days. This report investigates the role of feature enrichment, heat-aware weighting, and model families in the performance of models on both hot days and all days within a 24-hour prediction window. Our experiments resulted in three main findings. Firstly, we found that feature enrichment by including lagging, rolling statistics, and calendar features helped improve model accuracy. Secondly, we found that heat-aware cost weightings did improve accuracy and robustness of models on hot days without compromising overall accuracy of the model. Finally, we found that the sequence models, such as Long Short-Term Memory (LSTM) and Temporal Convolutional Networks (TCN), were more robust compared to baseline boosting models such as eXtreme Gradient Boost (XGBoost) and Light Gradient-Boosting Machine (LightGBM).

1. Introduction

Heatwaves push electricity demand to the edge: households and businesses lean on air-conditioning, straining feeders and the grid close to their operational limits. A sudden spike in consumption can force system operators to balance supply while avoiding load shedding and price shocks. Getting the demand forecast right on such days is therefore a reliability problem, critical for maintaining stability, preventing blackouts, and safeguarding market operations (Santagata et al., 2017; Sergio et al., 2025).

Reliable short-term electricity demand forecasting is central to secure and cost-efficient power system operation. Yet forecast accuracy tends to deteriorate precisely when accuracy matters most—during extreme weather events. This difficulty arises from two factors: (i) data imbalance, as extreme days are rare in historical records, and (ii) regime shifts, where demand–temperature relationships become nonlinear under heat stress. These challenges highlight the need for models that are not only accurate on average but also resilient under rare, high-impact conditions.

Neural sequence models such as Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) and Temporal Convolutional Networks (TCN; Bai et al., 2018) have advanced load forecasting by directly modelling long-range temporal dependencies and capturing non-linear demand–weather dynamics. Numerous studies have demonstrated the effectiveness of LSTMs for short-term electricity demand forecasting, showing that they outperform classical statistical models under conditions of strong seasonality and exogenous drivers (Khafaf et al., 2019; Bountiful et al., 2018). TCNs, meanwhile, offer a convolutional alternative that enables efficient parallel training while maintaining the ability to capture sharp peaks or regime shifts, and recent work has shown that TCNs can rival or even surpass LSTMs in multi-horizon load forecasting (Hamer et al., 2023). These strengths make sequence models particularly relevant for evaluating whether temporal-context modelling offers greater resilience under extreme heat conditions compared to boosting methods.

In contrast, gradient-boosted decision trees (GBDTs) such as XGBoost (Chen & Guestrin, 2016) and Light Gradient-Boosting Machine (LightGBM) (Ke et al., 2017) are attractive for electric load forecasting because they efficiently exploit heterogeneous tabular inputs such as lagged demand, rolling statistics, temperature extremes, and calendar indicators. They offer computational efficiency and interpretability (Hong & Fan, 2016), making them practical in operational settings. Moreover, their ensemble structure allows partitioning of the feature space adaptively, capturing nonlinear demand–temperature relationships and calendar effects without explicit temporal recursion, which further supports their suitability for demand forecasting under extreme weather conditions.

Feature engineering plays a critical role in electricity demand forecasting, particularly for improving baseline accuracy before addressing extreme events. Khafaf et al. (2019) demonstrated that combining weather and temporal features with demand history substantially improved day-ahead LSTM forecasts on Australian smart-meter data, achieving around 3% MAPE. Similarly, Bouktif et al. (2018) showed that optimising feature sets with genetic algorithms for load forecasting in France, alongside careful hyperparameter tuning, significantly enhanced LSTM accuracy compared to statistical and other machine learning baselines. These studies highlight that

systematically varying and enriching feature sets—through the inclusion of temporal lags, weather variables, and derived predictors—is a powerful way to strengthen forecasting performance and provide a robust foundation for modelling under more challenging extreme-weather conditions. Feature enrichment is especially valuable in such scenarios: incorporating maximum and minimum daily temperatures, or extreme-value statistics of recent demand, has been shown to reduce bias during hot days. However, the literature is still limited in systematically quantifying how different model families (e.g., boosting vs. sequence models) benefit from feature enrichment under extremes. Addressing this limitation is a central focus of our study.

Despite their strengths, both boosting and sequence models remain vulnerable to underestimation on extreme days. Recent work has attempted to mitigate this through loss modifications such as Extreme Value Loss (Ding et al., 2019) and event-weighted training strategies (e.g., SaDI framework; Liu et al., 2023), which up-weight rare but operationally critical samples such as heat waves. Building on this line of work, our study introduces a heat-aware reweighting scheme that explicitly increases the training cost of errors during hot days.

In summary, three strands of literature inform our study. First, feature engineering improves accuracy, but its effect on extreme-day robustness remains underexplored. Second, loss modifications such as Extreme Value Loss and event-triggered loss functions used in the SaDI framework have been shown to enhance rare-event performance, yet few studies have adapted these approaches specifically to electricity demand forecasting. Third, model families differ in strengths: boosting methods excel with heterogeneous features, while sequence models capture long temporal context—but few controlled studies directly compare their resilience to extreme events under identical conditions. Our work fills this gap by systematically evaluating feature enrichment, heat-aware loss weighting, and model-family contrasts in a unified experimental framework. This framework allows us to assess both forecast accuracy and robustness under heat stress, providing new evidence on how modelling choices influence resilience during extreme weather events.

Specifically, we pose three research questions:

RQ1. Do richer known-in-advance features (e.g., weather, calendar indicators, lagged demand) improve overall forecast accuracy?

RQ2. Does heat-aware reweighting reduce underestimation on hot days without degrading normal-day accuracy?

RQ3. Which model family is most robust under hot-day conditions — sequence models (LSTM, TCN) or boosting methods (LightGBM, XGBoost)?

Our contributions are threefold. First, we design and evaluate *Heat-Aware Cost Weighting (HACW)*, a simple yet effective training modification that explicitly up-weights hot days or hot slots to mitigate underestimation bias. Second, we conduct a controlled comparison of sequence and boosting families under identical, leakage-safe splits, with systematic feature ablations to quantify gains in both overall accuracy and robustness during hot days. Third, we perform a

horizon-wise error and bias analysis, showing that moderate reweighting improves forecasts during extreme heat with minimal disruption to normal-day performance.

2. Literature Review

2.1 Loss Modifications and Event-Weighted Training

A line of research has focused on loss modifications to improve robustness under rare events. Ding et al. (2019) introduced an Extreme Value Loss (EVL), derived from Extreme Value Theory, to better capture tail events in time series prediction. Their work demonstrated that explicitly weighting extremes within the training objective improves rare-event accuracy.

Building on this idea, Liu et al. (2023) proposed the SaDI framework, which uses event-triggered loss functions to up-weight samples influenced by extreme weather. Although not specific to LSTMs, SaDI highlights the broader benefit of sample-dependent loss functions in imbalance regimes, showing that reweighting rare-event samples improves robustness without major sacrifices on normal conditions.

Our study extends this tradition with *Heat-Aware Cost Weighting (HACW)*: a simple training modification that increases the loss penalty for errors on hot days and hot slots. Conceptually aligned with EVL and SaDI but tailored to the electricity demand forecasting context, HACW allows us to evaluate whether cost-sensitive training reduces underestimation bias during heat waves while maintaining accuracy on normal days. By evaluating HACW across different α values and both day-level and slot-level variants, we directly address RQ2: whether loss reweighting reduces underestimation bias during extreme heat without degrading performance on normal days.

2.2 Sequence Models (LSTM and TCN)

Neural sequence models have been widely adopted in energy forecasting due to their ability to capture long-range temporal dependencies and non-linear demand–weather dynamics. Unlike boosting methods that rely on explicitly engineered features, sequence models directly learn from temporal structure, making them particularly suitable for electricity demand forecasting where periodicity, seasonality, and regime shifts strongly influence load behaviour.

LSTM networks (Hochreiter & Schmidhuber, 1997) remain a cornerstone of sequence modelling, overcoming vanishing gradient problems through gating mechanisms that enable learning of multi-scale temporal patterns. In load forecasting, LSTMs have demonstrated strong performance: Khafaf et al. (2019) achieved $\sim 3\%$ MAPE on Australian smart-meter data using weather and temporal features, while Bouktif et al. (2018) showed that careful feature engineering combined with LSTMs significantly improved forecasting accuracy in France. More recent work has focused on robustness under extreme conditions. In parallel, Hamer et al. (2023) developed a forecast-aware LSTM for bias correction in air quality forecasting, combining clustering with a bi-directional LSTM to improve performance under extreme pollution events. This highlights the potential of LSTMs to be adapted through architectural and training innovations that improve resilience under rare but operationally critical extremes.

Temporal Convolutional Networks (TCNs) have emerged as a strong alternative to recurrent models for sequence modelling, using causal, dilated convolutions to capture long-range dependencies while enabling efficient parallelisation (Bai et al., 2018). Unlike LSTMs, which propagate sequentially, TCNs can process long input windows more efficiently, making them attractive for load forecasting tasks with pronounced periodicity. Their receptive fields can be tuned explicitly, offering flexibility in modelling both smooth seasonal patterns and sudden demand shifts. Comparative studies (Shaikh et al., 2023; Lara-Benítez et al., 2020) show that TCNs can perform competitively with, or even surpass, LSTMs in multi-horizon electricity demand forecasting, particularly under conditions of sharp peaks or regime shifts. These characteristics make TCNs a valuable benchmark alongside LSTMs for evaluating robustness during extreme weather conditions.

By benchmarking LSTM, we evaluate whether resilience during extreme heat is better achieved through temporal-context modelling or through loss modifications applied during training. In our setup, LSTM models are systematically tested under gradual feature enrichment (**RQ1**) and with heat-aware cost weighting to assess their individual contributions (**RQ2**). Once the most effective feature set was established, we applied the full feature set directly to TCNs, which serve as a complementary non-recurrent sequence model. This design allows us to isolate the effect of training modifications while providing a balanced comparison across model families (**RQ3**).

2.3 Gradient Boosted Decision Trees (XGBoost and LightGBM)

Gradient-boosted decision trees (GBDTs) such as XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) are widely used in load forecasting due to their ability to efficiently handle heterogeneous tabular features. Their advantages of speed and interpretability make them practical choices for operational demand forecasting.

XGBoost applies tree-based ensemble learning with gradient boosting and has been successfully applied in electricity demand forecasting across diverse use cases, including residential consumption (Dong et al., 2023) and aggregate load forecasting (Mohamad et al., 2023). Different preprocessing methods have been explored to enhance XGBoost performance, such as principal component analysis (Mohamed et al., 2023) and window-based multi-output regression frameworks (Elsayed et al., 2021), with results showing strong performance relative to deep neural networks. Hybrid approaches have also been proposed, for example combining LSTMs with XGBoost to integrate sequential modelling with error correction (Ajder, Hamza & Ayaz, 2025). These studies collectively demonstrate the robustness and versatility of XGBoost in capturing complex feature–demand relationships for short-term forecasting.

LightGBM (Ke et al., 2017) is a tree-based gradient boosting framework developed by Microsoft that has been widely adopted for predictive modelling tasks requiring both accuracy and training efficiency. It has been successfully applied across meteorology and energy forecasting, including rainfall prediction (Malakouti, 2023), air-quality and pollution modelling (Abeebhajid & Obagbuwa, 2024), and short-term electricity demand forecasting (Rubattu et al., 2023). Studies emphasise LightGBM’s ability to capture non-linear feature interactions and temporal dependencies through feature engineering (lagged demand, weather variables, calendar indicators) combined with advanced hyperparameter tuning. Notably, Rubattu et al. (2023) showed that

LightGBM with leaf-wise growth can outperform XGBoost in RMSE and probabilistic metrics, especially under temperature extremes where sharp demand spikes occur, highlighting its suitability for forecasting rare but operationally critical events such as extreme temperatures and severe storms.

Given their demonstrated strengths and limitations, we include LightGBM and XGBoost as boosting baselines in RQ3, providing a direct contrast to sequence models (LSTM and TCN) and enabling us to test which model family offers greater robustness under extreme heat.

3. Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to understand the characteristics of the electricity demand, temperature, and forecast datasets, and to identify how extreme weather conditions affect forecasting accuracy. This stage provided both descriptive insights and empirical motivation for our modelling design.

To prepare the datasets, DATETIME fields were standardised to a uniform format and time zone (UTC), and timestamps were rounded to the nearest half-hour to align with demand and forecast records. Temporal variables (year, month, day, day-of-week) were extracted for later use as features.

We next checked for missing values and produced descriptive summaries of each column. Temperature distributions were explored using boxplots, which revealed clear seasonal patterns and highlighted the presence of both hot and cold extremes. To focus on weather conditions relevant to our research question, we defined thresholds for extreme days as below 5 °C (cold) and above 30 °C (hot). This enabled us to create an “extreme-day table” capturing minimum and maximum temperatures for flagged days, providing the foundation for later analysis of forecasting under normal versus extreme conditions.

After cleaning, the three datasets (actual demand, forecast demand, and temperature) were merged into a single half-hourly dataset. For each half-hour interval, the final dataset contains the actual demand, the forecast demand, and the corresponding temperature. To ensure consistency and avoid duplication from multiple update runs, we retained only the most recent forecast available at each interval for the exploratory data analysis. This consolidated structure allowed us to directly link forecast errors with temperature variations.

3.1 Daily temperature distribution

The daily maximum and minimum temperatures demonstrate the seasonal clustering of extremes. As shown in Figure 1, monthly boxplots of half-hourly temperatures show a clear seasonal cycle in New South Wales: median winter temperatures range from 11–15 °C (June–August), while summer medians reach 21–23 °C (December–February). Hot days (≥ 30 °C) occur almost exclusively in summer, with outliers above 40 °C concentrated in January and February. Cold days (≤ 5 °C) occur in winter, with occasional sub-zero minimums in June and July.

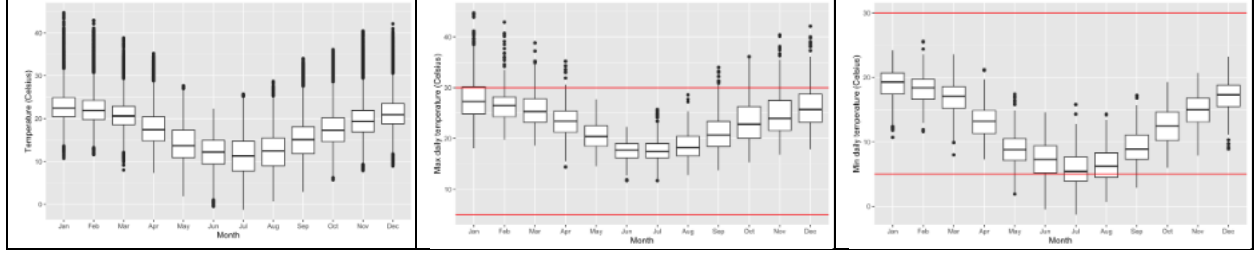


Figure 1. Monthly temperature distributions in NSW: (*left*) half-hourly values. (*centre*) daily maximum with hot threshold (≥ 30 °C, red line), and (*right*) daily minima with cold threshold (≤ 5 °C, red line).

Initially, a threshold of 35 °C was considered for defining hot days, but exploratory analysis showed that forecast errors began to rise substantially beyond 30 °C. Therefore, 30 °C was adopted as the threshold for hot extremes. These findings confirm that the dataset captures both seasonal cycles and extreme events, providing a robust basis for analysing demand–temperature interactions under stress.

3.2 Relationship between temperature and demand

Linking temperature and demand at the daily scale revealed that while most days fall within a moderate range (5–30 °C), demand volatility increases sharply on hot days. Scatterplots of daily maximum and minimum temperatures against total daily demand (see Figure 2) reveal that extremes occupy only a small fraction of the dataset. Out of 4,092 days, 355 (8.6%) were classified as cold (≤ 5 °C), 343 (8.4%) as hot (≥ 30 °C), and the remaining 3,661 (82.9%) as normal. This imbalance highlights that while most days provide “normal” training data, rare hot days contribute disproportionately to volatility and sharp demand spikes. This confirms that demand volatility increases sharply at higher temperatures, justifying the need to focus on extreme-day forecasting.

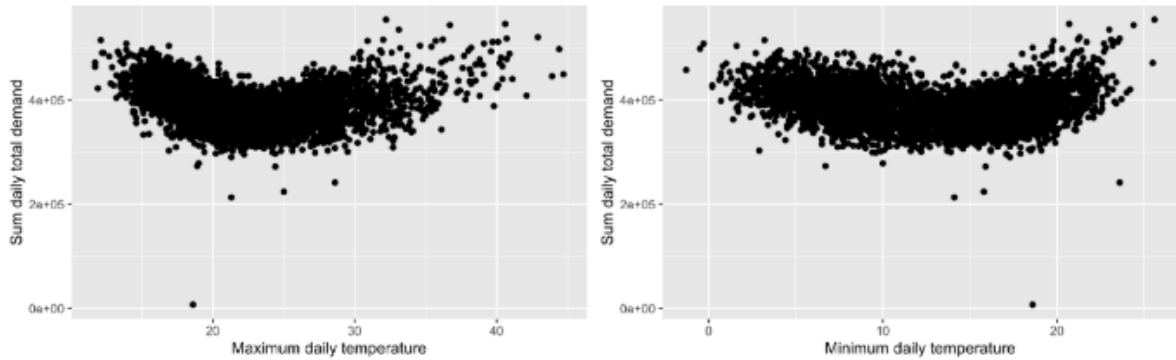


Figure 2. Relationship between daily electricity demand and temperature: (*left*) demand vs. daily maximum temperature and (*right*) demand vs. daily minimum temperature, showing heightened volatility under hot extremes (≥ 30 °C).

3.3 Hot-day characteristics

To evaluate how forecasting accuracy changes under extreme heat, we created a consolidated half-hourly dataset combining actual demand, forecast demand, and temperature, and computed forecast errors (actual – forecast). The forecast demand closest to the interval time was selected as representative of the best estimate from the current forecast model. Each interval was labelled as Hot, Cold, or Normal using the same thresholds defined earlier (≤ 5 °C for cold, ≥ 30 °C for hot). This allowed us to directly analyse systematic differences in forecast performance across weather regimes.

Scatter plots of actual versus forecast demand (see Figure 3) show a strong overall correlation, confirming that forecasts capture broad demand dynamics. However, when stratified by temperature, deviations widen under hot conditions. Cold days remain tightly clustered around the diagonal, while hot days exhibit larger dispersion, indicating that extreme heat is associated with systematic underestimation.

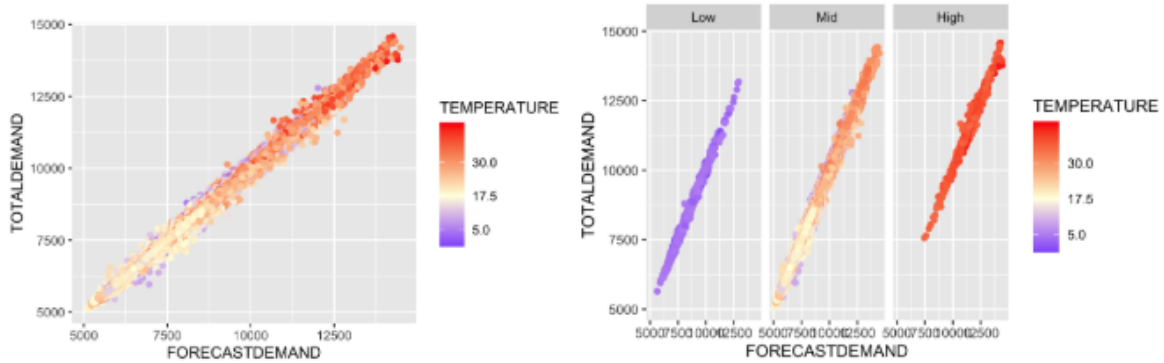


Figure 3. Forecast demand vs. actual demand, coloured by temperature: *(left)* overall relationship shows strong correlation but wider deviations at higher temperatures; *(right)* stratified plots for low, mid, and high temperature ranges, highlighting larger errors under hot conditions.

Error profiles by temperature bin (Figure 4) further highlight this pattern. Mean Absolute Error (MAE) is relatively stable from -5 °C to 25 °C but rises sharply beyond 30 °C, while RMSE shows an even steeper increase, particularly in the hottest bins (≥ 35 °C). Whereas Normal (5 – 30 °C) and Cold (≤ 5 °C) days both exhibit RMSE around 80 – 85 MW, Hot days (≥ 30 °C) reach ~ 130 MW, more than 1.5 times higher. This demonstrates that extreme heat introduces substantially larger and more volatile forecast errors, consistent with demand spikes from cooling loads. The higher forecast errors on hot days motivates the selection of hot weather days as the focus of this project.

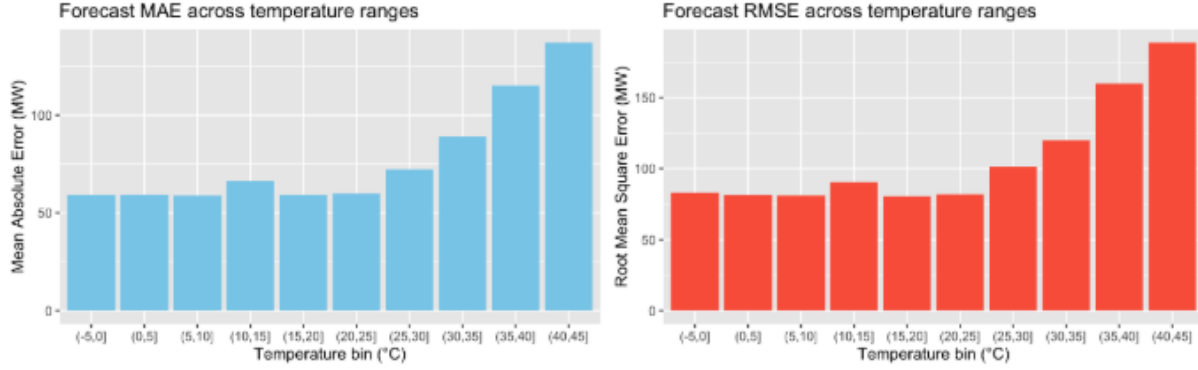


Figure 4. Forecast error across temperature ranges: (*left*) Mean Absolute Error (MAE) and (*right*) Root Mean Square Error (RMSE). Both metrics rise notably at high temperatures, with RMSE showing the steepest deterioration.

Finally, examining MAE during the timeslots below and above our extreme thresholds (Figure 5) show that higher errors often occur during the non-extreme temperature portions of the day. Most points for cold slots (timeslots where temperature $\leq 5^\circ\text{C}$) cluster close to or below the diagonal, showing forecast errors remain stable in the cold slots. In contrast, hot extreme slots—especially those exceeding 40°C —show substantially higher MAE, with many points lying well above the diagonal. These results establish that extreme heat is one of the most critical regimes for assessing and improving electricity demand forecasting accuracy, particularly across the entire day and not during the hot slot.

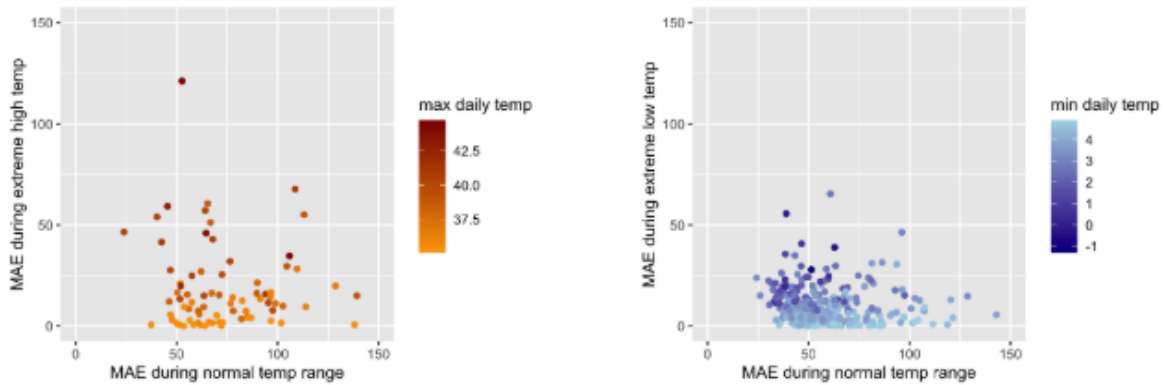


Figure 5. Comparison of daily Mean Absolute Error (MAE) between normal and extreme conditions within the same day: (*left*) hot extremes, showing larger deviations especially above 40°C , and (*right*) cold extremes, where errors remain relatively stable near the diagonal.

Together, these findings underscore the need for temperature-aware features and weighting strategies in subsequent modelling over the course of hot days. In particular, incorporating feature engineering (e.g., lagged temperature indicators) and applying targeted cost-weighting for extreme days can help mitigate the systematic underestimation observed during hot conditions.

4. Data and Pre-processing

This section describes how the three raw data sources—electricity demand, temperature, and forecast demand—were cleaned, aligned, and transformed into a modelling dataset. We document each step from timestamp alignment to feature construction, and we explain the choices made to ensure leakage-safe splits, fair evaluation, and consistency with the exploratory findings presented earlier.

4.1 Data sources

Our study draws on half-hourly electricity demand for New South Wales obtained from the Australian Energy Market Operator (AEMO), pre-dispatch forecast demand from AEMO’s Market Management System, and air temperature from the Bankstown Airport weather station (Australian Data Archive for Meteorology). The dataset spans January 2010 to March 2021, covering more than a decade of diverse weather conditions. To control for calendar effects, we additionally included public holiday flags and weekday/weekend indicators for New South Wales. All timestamps were standardised to UTC to avoid complications from daylight saving time.

4.2 Timestamp alignment and merging

The three raw datasets were merged into a common half-hour grid. For temperature, raw readings were rounded to the nearest 30 minutes and averaged within each slot; small gaps of up to three hours were filled by interpolation. Actual demand was similarly rounded and averaged within duplicated slots. For forecast demand, multiple issue times exist for each target interval; to replicate the “latest available” information at any point in time, we retained only the most recent forecast issued for each 30-minute slot. The three sources were then merged into a unified table with columns for actual demand, forecast demand, and temperature, along with basic time features (hour of day, day of week, and weekend/holiday flags). This process yielded a complete sequence of ~196,000 half-hour records with no missing values in the base columns.

4.3 Feature engineering

Feature engineering was a critical step in capturing the temporal dynamics of electricity demand and its sensitivity to extreme weather. As shown in Table 1, we focused on designing lagged features, rolling statistics, and temperature indicators to reflect both short-term persistence and longer seasonal cycles.

Lag features capture natural daily and weekly rhythms in demand. For instance, demand 24 hours earlier ($y_{\text{lag}48}$) reflects typical daily cycles, while $y_{\text{lag}336}$ (7 days ago) encodes weekly usage patterns. These are particularly valuable because extreme days often follow similar structures across consecutive days. Short lags ($y_{\text{lag}1}$) capture immediate inertia, where a sudden jump in load tends to persist into the next interval.

Rolling features smooth demand and temperature over daily or weekly windows, providing baselines against which the model can detect unusual deviations. For example, the 24-hour rolling

standard deviation (`y_roll_std_48`) highlights recent volatility, often elevated during hot spells. Likewise, rolling maxima and minima for temperature capture the build-up of heatwaves or cold snaps, which single observations may miss.

Table 1. Summary of engineered features

Feature type	Feature name	Description
Demand history	<code>y_lag1</code>	Demand 30 mins ago (short-term inertia).
	<code>y_lag48</code>	Demand at the same time yesterday (daily cycle).
	<code>y_lag336</code>	Demand at the same time last week (weekly cycle).
Demand rolling	<code>y_roll_mean_48</code>	Average demand over past 24h (short-term baseline).
	<code>y_roll_std_48</code>	Demand volatility over past 24h (recent variability).
	<code>y_roll_mean_336</code>	Average demand over past 7 days (weekly baseline).
Temperature lags	<code>temp_lag1~3</code>	Temperatures 30~90 mins ago (short-term persistence).
	<code>temp_lag48</code>	Temperatures at the same time yesterday (daily cycle).
Temperature rolling	<code>temp_roll_max_48</code>	Maximum temperature in past 24h (heat build-up indicator).
	<code>temp_roll_min_48</code>	Minimum temperature in past 24h (cold spell indicator).

Temperature lags were included to reflect persistence in heating and cooling loads. For example, yesterday’s temperature at the same time (`temp_lag48`) complements the daily demand lag, while short lags (30–90 minutes) encode the thermal inertia of households and buildings. These features allow the model to anticipate demand surges driven by extended hot or cold periods rather than isolated values.

Together, this feature design aims to improve robustness during extreme conditions by explicitly modelling temporal persistence and volatility.

4.4 Data cleaning and splitting

After merging, we verified both structural completeness and internal consistency. First, all records contained valid demand and temperature values, with missing values only arising later from lag or rolling operations. These leading NAs were safely dropped prior to modelling. Second, we confirmed monotone time coverage across the entire window and applied filters to ensure that each evaluation day contained a full set of 48 half-hour observations. Importantly, we kept all extreme values (e.g., very hot days above 40 °C) in the dataset, rather than trimming or reducing them, because these rare events are precisely the focus of our study. Outliers such as days above 40 °C were preserved as they represent the operationally critical conditions under which forecasting accuracy most strongly deteriorates.

To prevent leakage and reflect operational forecasting, the dataset was split chronologically into non-overlapping periods: training (Jan 2010 – Dec 2017), validation (Jan 2018 – Dec 2018), and

test (Jan 2019 – Mar 2021). This split ensures that models are trained on historical conditions, tuned on an intermediate year, and evaluated on the most recent period containing multiple extreme-heat events unseen during training.

4.5 Evaluation cohorts

For consistent evaluation across all models, we defined two fixed cohorts from the test split. The *Overall-all* cohort includes every complete day in the test period (805 days, or 38,640 half-hour slots), representing average operating conditions. The *Hotday-all* cohort is a targeted subset containing only the 80 days where the hot-day flag is active (≥ 30 °C daily maximum), yielding 3,840 half-hour slots. This design allows us to specifically stress-test models under heat extremes while retaining a baseline comparison against all conditions.

These cohorts serve complementary purposes: *Overall-all* assesses average model behaviour under everyday conditions, while *Hotday-all* isolates performance under extreme heat. Using both enables us to quantify not only statistical accuracy but also robustness and operational reliability, which are critical for deployment in industry settings.

4.6 Summary

These design choices align the data pipeline with our research questions. EDA confirmed that forecasting errors rise sharply on hot days, motivating the inclusion of temperature lags, extreme-day flags, and calendar features—directly addressing **RQ1** (feature enrichment). Finally, the fixed cohorts ensure fair and reproducible evaluation, allowing us to quantify **RQ2** (impact of heat-aware weighting) and **RQ3** (model-family robustness) and under identical experimental conditions.

5. Methods

5.1 Experimental Design

Our methodology was designed to isolate the contributions of feature enrichment (RQ1), heat-aware reweighting (RQ2), and model family evaluation (RQ3) under identical conditions. All models were trained on the chronological split described in Section 4.4, with evaluation conducted on the *Overall-all* and *Hotday-all* cohorts presented in Section 4.5. This setup ensures comparability across model families while providing stress-tests under extreme heat.

5.2 Sequence Models: LSTM and TCN

5.2.1 LSTM

We implemented a sequence-to-sequence LSTM to forecast 48 half-hour steps (one full day) ahead, conditioned on the preceding 48 half-hour slots. The network consists of a single LSTM layer (64 hidden units), followed by a dense layer to map hidden states to outputs. Training used Adam ($\text{lr}=0.001$), batch size 256, gradient clipping ($\text{norm}=1.0$), and early stopping ($\text{patience}=10$).

To address **RQ1**, LSTM was trained with progressively richer feature sets:

- *fs2_min*: demand + temperature
- *fs3_time*: adds time-of-day cycles
- *fs4_cal*: adds calendar indicators
- *fs_strong*: full set (lags, rolling stats, calendar)

To address **RQ2**, we implemented *Heat-Aware Cost Weighting (HACW)*. Training samples flagged as hot days or hot slots were up-weighted ($\alpha \in \{1.5, 2.0, 3.0\}$). This weighting encourages the model to prioritise accuracy on high-risk periods without discarding performance under normal conditions. The baseline model uses uniform weights ($\alpha = 1.0$).

5.2.2 TCN

To benchmark non-recurrent sequence models for **RQ3**, we implemented a Temporal Convolutional Network comprising five residual blocks, each containing a 1-D dilated convolutional layer (kernel size = 5), followed by batch normalisation, ReLU activation, and dropout. Each residual block included a skip connection from the block input to its output. The output channels began at 32 and the dilation rate at 1, both doubling with each successive block to expand the receptive field efficiently. A final 1×1 convolutional layer produced the 48-step output, matching the forecasting horizon of other models. The architecture was trained using early stopping and learning-rate reduction callbacks to maintain training stability.

5.3 Boosting Models: XGBoost and LightGBM

5.3.1 XGBoost

To evaluate the performance of boosting-based methods for **RQ3**, we implemented a multi-output XGBoost model for 48-step forecasting, following Elsayed et al. (2021). As shown in Figure 6, the model adopts a window-based configuration that restructures sequential inputs into lagged windows. Each training sample consists of flattened sequences of previous target values (demand) and covariates (temperature, calendar indicators, and rolling statistics). The sklearn.multioutput wrapper was used to generate multiple output horizons, while the xgboost package handled model training and prediction.

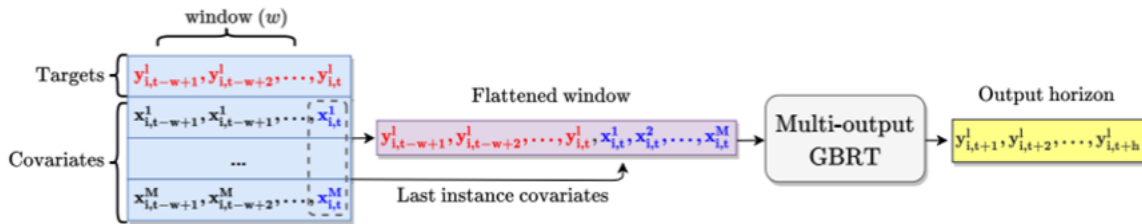


Figure 6. Reconfiguration of a single window of input for multi-output forecasting (Elsayed et al., 2021).

All lagged demand and temperature features were scaled between -0.5 and 0.5 for numerical stability. The multi-output setup ensures the model predicts all 48 half-hour steps simultaneously, maintaining temporal consistency across the forecast horizon.

Hyperparameters were adapted from prior energy forecasting studies (Elsayed et al., 2021) and applied without cross-validation due to time constraints. Key parameters were: *learning_rate* = 0.2 , *n_estimators* = 20 , *max_depth* = 8 , *min_child_weight* = 1 , *gamma* = 0.0 , *subsample* = 0.8 , *colsample_bytree* = 0.8 , and *scale_pos_weight* = 1 . These values were chosen to balance complexity and generalisation, providing stable convergence across forecast intervals.

5.3.2 LightGBM

To further evaluate boosting-based methods for **RQ3**, we implemented LightGBM as a fast, industry-oriented baseline for multi-horizon load forecasting. LightGBM’s histogram-based splits and leaf-wise growth enable strong accuracy at low computational cost, making it well-suited for large-scale tabular datasets (Ke et al., 2017).

Inputs mirrored those used for XGBoost—lagged demand and temperature features, rolling statistics, and calendar indicators—while training remained leakage-free by conditioning strictly on timestamps preceding the test period. To efficiently cover the 48-step horizon, the model was trained in a chunked $48 \rightarrow 48$ setup, comprising four 12-step multi-output models spanning horizons $t+1-12$, $13-24$, $25-36$, and $37-48$. This design preserves horizon coverage while keeping runtime modest (Ke et al., 2017; Hong and Fan, 2016).

Training was conducted using LightGBM’s gradient boosting framework with moderate tree depth and subsampling to maintain generalisation. Each model used identical data splits and feature inputs as XGBoost, ensuring a fair basis for model-family comparison.

6. Experiment Results

Model performance was evaluated using four standard error metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Bias. MAE and RMSE measure the magnitude of absolute and squared deviations, while MAPE expresses errors relative to actual demand levels in percentage terms. Bias quantifies systematic over- or under-estimation, indicating directional tendencies in model forecasts. These complementary metrics collectively capture both the magnitude and direction of forecast errors across all test cohorts and horizons. For detailed metric formulations and the slot-wise/horizon-wise aggregation procedure, see Appendix A.

6.1 RQ1 – Effect of Feature Enrichment (LSTM)

To evaluate **RQ1**, we analysed how progressively richer input features influence the baseline LSTM’s forecasting performance. As described in Section 5.2.1, four feature configurations were used, ranging from the minimal (*fs2_min*) setup (demand + temperature) to the full-feature model (*fs_strong*), which incorporates all temporal, calendar, and rolling features.

Table 2. Aggregate performance across feature configurations on the Overall-all cohort

Feature	MAE	RMSE	MAPE	Bias	Δ MAE	Δ RMSE	Δ MAPE	Δ Bias
fs_strong	325.36	468.02	4.12	1.97	0.00	0.00	0.00	0.00
fs4_cal	359.80	507.00	4.62	52.10	34.44	38.98	0.50	50.13
fs3_time	384.95	547.96	5.00	106.44	59.59	79.94	0.88	104.47
fs2_min	405.54	579.19	5.27	151.62	80.18	111.17	1.15	149.65

Table 2 summarises the aggregate performance across feature configurations. Error magnitudes decrease monotonically as additional temporal and calendar variables are introduced, confirming that feature enrichment enhances both accuracy and generalisation under normal operating conditions. The full-feature model (*fs_strong*) achieves the lowest MAE (325 MW) and RMSE (468 MW), outperforming the minimal feature setup (*fs2_min*) by approximately 20 %. These improvements are accompanied by a substantial reduction in MAPE and Bias, indicating not only higher predictive accuracy but also reduced systematic error. The Δ -columns quantify performance improvements relative to *fs_strong*, with smaller values representing closer alignment to the optimal configuration.

The horizon-wise evaluation further reinforces these findings. As shown in Figure 7, error magnitudes increase with lead time, as expected in short-term forecasting; however, *fs_strong* consistently yields the flattest curves across horizons, indicating slower error growth and superior long-range stability. In contrast, *fs2_min* systematically over-predicts demand, with larger bias and higher MAPE beyond 12–24 hours. The inclusion of calendar and rolling statistics notably mitigates this drift, producing more stable and unbiased forecasts.

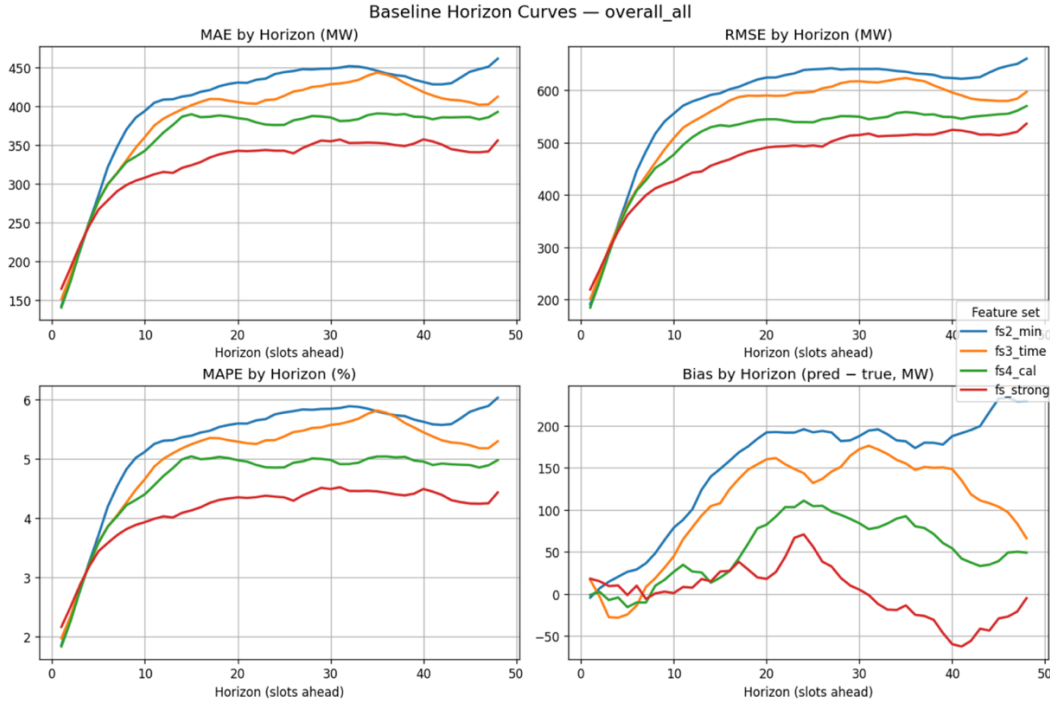


Figure 7. Performance of progressively enriched LSTM feature sets: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

In summary, the *fs_strong* configuration emerges as the most effective baseline, providing the most accurate and stable forecasts under normal operating conditions. This model is therefore adopted as the reference configuration for subsequent experiments incorporating hot-day and hot-slot weighting schemes (addressing RQ2).

6.2 RQ2 – Impact of Head-Aware Cost Weighting (LSTM)

To address **RQ2**, we investigated whether *Heat-Aware Cost Weighting (HACW)* improves LSTM forecast robustness under extreme heat while maintaining accuracy on normal days. Two complementary weighting schemes were tested:

- **Hot-day weighting** – assigning higher weights to entire hot-day samples.
- **Hot-slot weighting** – up-weighting only the peak demand hours within hot days.

6.2.1 Method Overview

The *fs_strong* feature configuration (Section 6.1) was used as the baseline model. Training samples flagged as hot days or hot slots were re-weighted by $\alpha \in \{1.5, 2.0, 3.0\}$. This re-weighting forces the model to prioritise accuracy on high-stress periods while preserving stability on normal days.

6.2.2 Results - Hot-Day Weighting

The objective is twofold:

1. Evaluate whether weighting preserves performance on *overall* (all-day) conditions.
2. Determine whether weighting improves accuracy on *hot days* where forecasting errors are most critical.

(a) Overall Cohort Results

As shown in Table 3, the *fs_strong* baseline remains the most accurate overall (MAE = 325 MW, RMSE = 468 MW). Weighted models incur only minor penalties (+2–10 MW). The $\alpha = 1.5$ variant slightly over-corrects bias (+65 MW), whereas $\alpha = 3.0$ slightly reduces bias but inflates RMSE. Overall, the penalty for weighting is minimal (< 3%) and performance remains stable across horizons.

Table 3. Performance of hot-day weighted LSTM models on the overall cohort

Feature	MAE	RMSE	MAPE	Bias	Δ MAE	Δ RMSE	Δ MAPE	Δ Bias
fs_strong	325.36	468.02	4.12	1.97	0.00	0.00	0.00	0.00
hot_day_a1.5	335.00	477.20	4.29	66.62	9.64	9.18	0.17	64.65
hot_day_a2.0	333.19	475.14	4.23	1.57	7.83	7.12	0.11	-0.4
hot_day_a3.0	327.57	472.65	4.14	-11.61	2.21	4.63	0.02	-13.58

Figure 8 plots confirm this:

- *fs_strong* maintains lower MAE and RMSE at short–medium horizons,
- Weighted variants converge near the baseline for longer horizons,
- MAPE differences remain within 0.2 %.

These results confirm that hot-day weighting introduces negligible degradation under normal conditions.

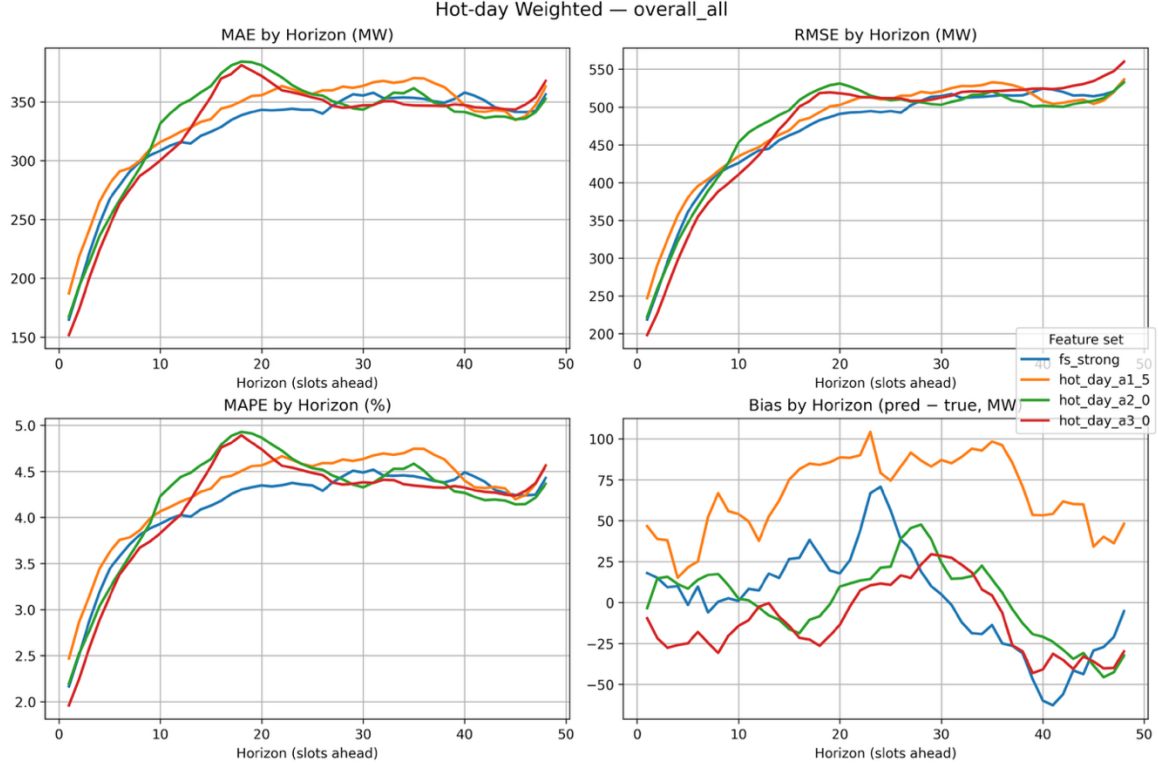


Figure 8. Overall cohort performance of hot-day weighted variants: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

(b) Hot-Day Cohort results

As shown in Table 4, the $\alpha = 1.5$ model achieves the **lowest MAE (630 MW)** and **RMSE (848 MW)**, outperforming *fs_strong* by ~ 45 MW and 73 MW respectively. Heavier weights ($\alpha = 2.0, 3.0$) further reduce bias but increase overall errors, indicating over-correction.

Table 4. Performance of hot-day weighted LSTM models on the hot-day cohort

Feature	MAE	RMSE	MAPE	Bias	Δ MAE	Δ RMSE	Δ MAPE	Δ Bias
<i>fs_strong</i>	675.39	921.19	7.48	-335.69	0.00	0.00	0.00	0.00
hot_day_a1.5	630.12	847.7	7.06	-250.37	-45.27	-73.49	-0.42	85.32
hot_day_a2.0	649.45	878.57	7.18	-339.60	-25.94	-42.62	-0.30	-3.91
hot_day_a3.0	684.27	943.74	7.52	-401.68	8.88	22.55	0.04	-66.00

Figure 9 shows that $\alpha = 1.5$ consistently tracks below fs_strong across all horizons, maintaining lower bias and error magnitudes.

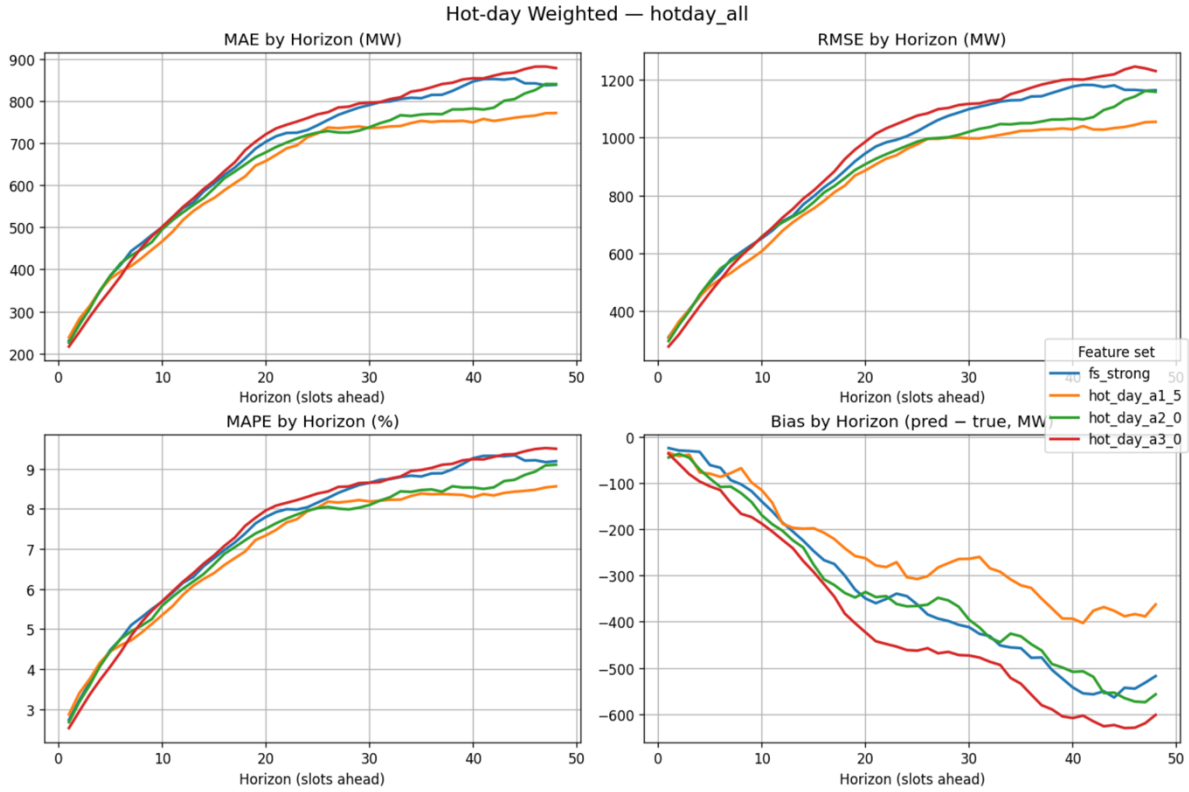


Figure 9. Hot-day cohort performance of hot-day weighted variants: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

Conclusion (Hot-Day Weighting):

In conclusion, the results indicate that hot-day weighting does not yield gains on the overall cohort but becomes valuable under hot-day stress conditions. Among the variants, $\alpha=1.5$ consistently strikes the best balance, preserving accuracy on normal days while enhancing robustness on extreme days.

6.2.3 Results - Hot-Slot Weighting

(a) Overall Cohort Results

As demonstrated in Table 5, hot-slot weighting does not provide consistent improvements. All weighted variants show slightly higher MAE and RMSE, with $\alpha = 2.0$ performing closest to baseline.

Table 5. Performance of hot-slot weighted LSTM models on the overall cohort

Feature	MAE	RMSE	MAPE	Bias	Δ MAE	Δ RMSE	Δ MAPE	Δ Bias
fs_strong	325.36	468.02	4.12	1.97	0.00	0.00	0.00	0.00
hot_slot_a1.5	341.47	483.47	4.33	-7.25	16.11	15.45	0.21	-9.22
hot_slot_a2.0	325.63	467.70	4.14	15.13	0.27	-0.32	0.02	13.16
hot_slot_a3.0	329.30	478.38	4.18	4.98	3.94	10.36	0.06	3.01

The horizon plots show a clear trade-off: $\alpha = 1.5$ reduces bias but introduces larger variance and higher MAPE at mid-range horizons.

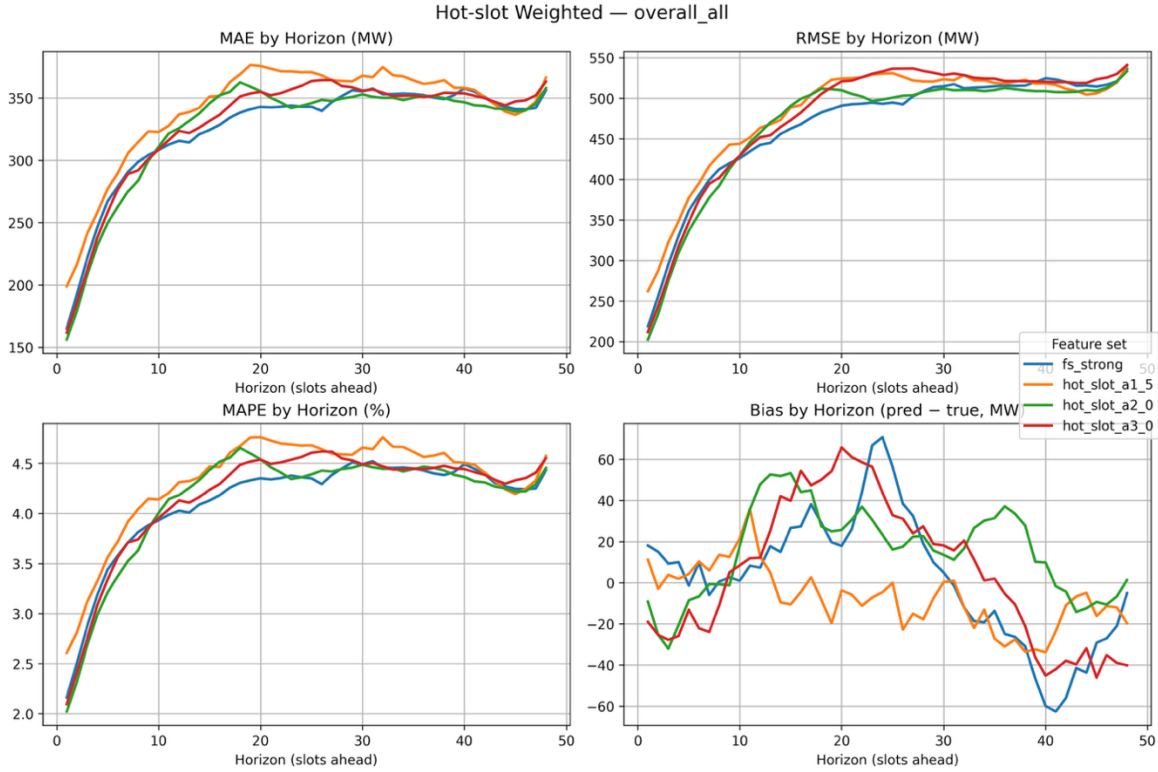


Figure 10. Overall cohort performance of hot-slot weighted variants: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

Overall, these results suggest that while hot-slot weighting can shift bias behaviour, it introduces higher errors on average. In contrast to hot-day weighting, which delivered targeted robustness on extreme days, hot-slot weighting appears less effective at balancing the trade-off between peak-hour accuracy and general performance.

(b) Hot-Day Cohort Results

As shown in Table 6, hot-slot weighting performs better when limited to extreme days. The $\alpha = 2.0$ variant achieves the best trade-off (MAE = 638 MW, RMSE = 864 MW), improving over baseline by ~ 37 MW and 57 MW respectively. This demonstrates that targeted up-weighting during peak stress periods improves local accuracy without harming full-day stability.

Table 6. Performance of hot-slot weighted LSTM models on the hot-day cohort

Feature	MAE	RMSE	MAPE	Bias	Δ MAE	Δ RMSE	Δ MAPE	Δ Bias
fs_strong	675.39	921.19	7.48	-335.69	0.00	0.00	0.00	0.00
hot_slot_a1.5	659.17	893.45	7.28	-369.71	-16.22	-27.74	-0.20	-34.02
hot_slot_a2.0	638.19	864.47	7.12	-289.62	-37.20	-56.72	-0.36	46.07
hot_slot_a3.0	659.08	903.23	7.34	-312.78	-16.31	-17.96	-0.14	22.91

Figure 11 demonstrates that $\alpha = 2.0$ consistently tracks below the baseline across all horizons, yielding lower MAE and RMSE while maintaining stable MAPE. Moderate weights ($\alpha = 1.5$ – 2.0) most effectively reduce underestimation during hot-day peaks.

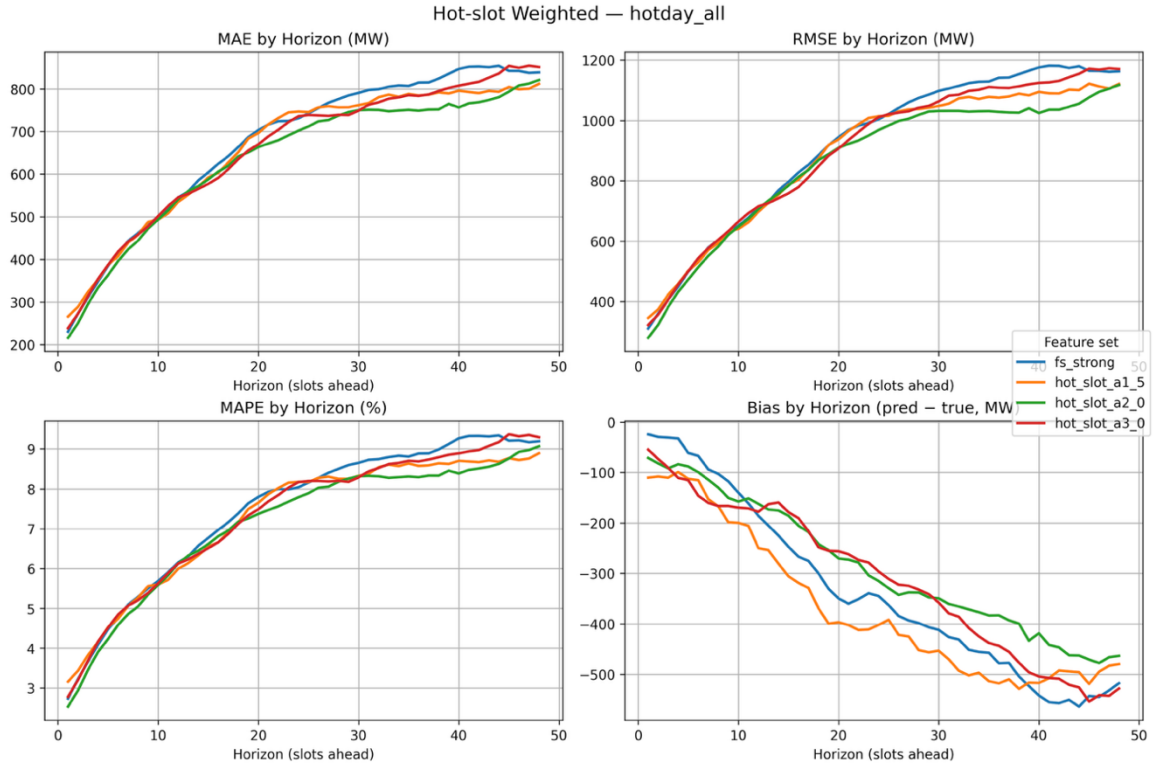


Figure 11. Hot-day cohort performance of hot-slot weighted variants: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

Conclusion (Hot-Slot Weighting):

Hot-slot weighting does not enhance overall accuracy but effectively strengthens robustness under extreme conditions. By prioritising learning on peak-demand hours, it mitigates underestimation and improves forecast reliability when system stress is highest, addressing RQ2’s aim of enhancing performance during critical periods.

6.2.4 Summary of RQ2 Findings

As shown in Table 7, the results demonstrate how **Heat-Aware Cost Weighting (HACW)** enhances model robustness during extreme-heat conditions through two complementary weighting strategies:

- **Hot-day weighting ($\alpha = 1.5$)** most effectively enhances *robustness* under heat stress, reducing large forecast deviations without harming baseline accuracy.
- **Hot-slot weighting ($\alpha = 2.0$)** sharpens *peak-hour accuracy* but introduces slightly higher variability under normal conditions.
- Overall, **moderate weighting** improves resilience while maintaining generalisation.

Table 7. Aggregate comparison of HACW variants (LSTM fs_strong baseline)

Feature	Overall MAE	Overall RMSE	Hotday MAE	Hotday RMSE	Comment
fs_strong	325.36	468.02	675.39	921.19	Baseline: accurate overall, underestimate hot-day peaks
hot_day_a1.5	335.00	477.20	630.12	847.70	Best overall trade-off: stronger hot-day robustness
hot_slot_a2.0	325.63	467.70	638.19	864.47	Best peak-hour focus: slightly less stable overall

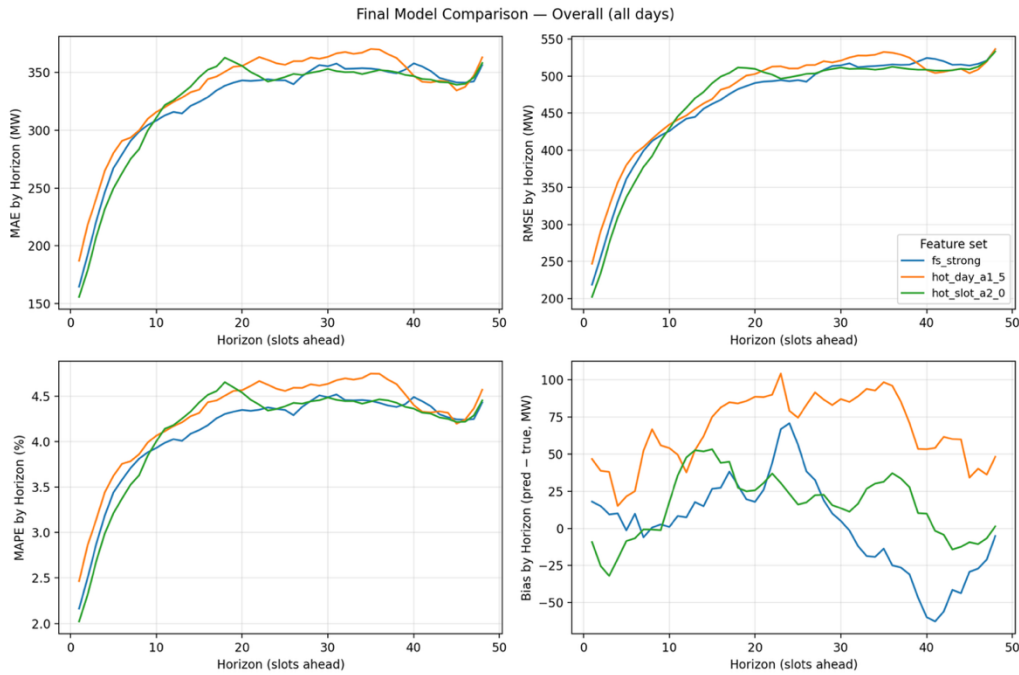


Figure 12. Overall cohort performance of HACW variants: (a) MAE, (b) RMSE, (c) MAPE, (d) Bias curves over 48 half-hour horizons

Figure 12 demonstrates that across the **overall cohort**, all three configurations show similar trajectories, confirming that HACW does not degrade general-day accuracy. The *hot_day_a1.5* model incurs only a minor (≈ 10 MW) increase in MAE and $< 0.2\%$ MAPE, while *hot_slot_a2.0* performs slightly better at mid-range horizons. Bias curves reveal that moderate weighting stabilises forecast deviations without amplifying variance, preserving day-to-day reliability.

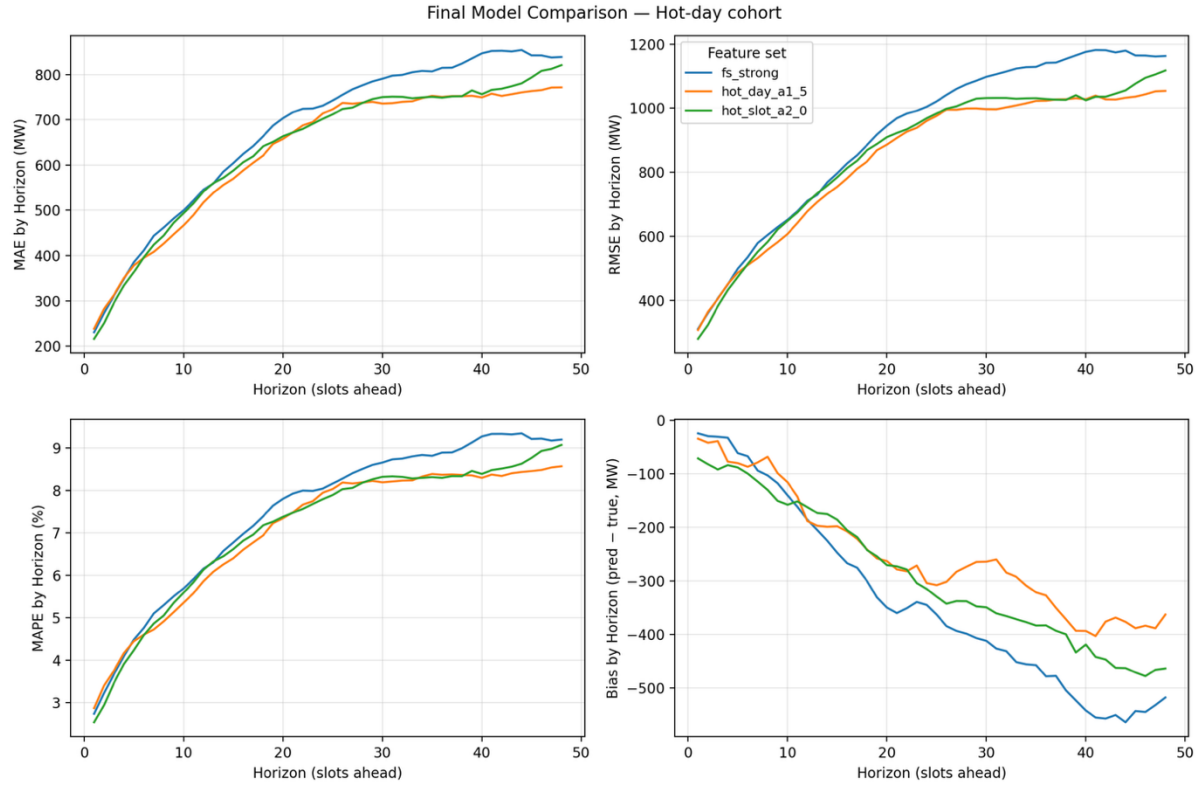


Figure 13. Hot-day cohort performance of HACW variants: (a) MAE, (b) RMSE, (c) MAPE, (c) Bias curves under extreme heat conditions

Figure 13 shows that for the hot-day cohort, clear gains emerge: both *hot_day_a1.5* and *hot_slot_a2.0* outperform the baseline *fs_strong*, reducing MAE by $\approx 45\text{--}70$ MW and RMSE by ≈ 70 MW. The *hot_day_a1.5* variant delivers the most stable improvements, lowering underestimation bias and maintaining tighter accuracy across horizons, whereas *hot_slot_a2.0* focuses on peak hours with slightly higher variability.

Conclusion (RQ2)

These experiments confirm that **moderate hot-aware weighting** substantially strengthens model reliability under extreme-heat conditions while preserving normal-day accuracy. Among the tested schemes, **hot_day_a1.5** achieves the most balanced performance — sustaining baseline precision during ordinary periods while markedly reducing large errors and bias during heatwaves. This configuration therefore provides the best balance of robustness and generalisability, representing a practical and operationally resilient model for real-world electricity-demand forecasting.

6.3 RQ3 – Model Family Evaluation (LSTM, TCN, XGBoost and Light GBM)

This section addresses **RQ3**, comparing sequence-based (LSTM, TCN) and boosting-based (XGBoost, LightGBM) models in forecasting accuracy and robustness across normal and extreme heat scenarios. The aim is to identify which model architecture generalises best under stress while maintaining baseline stability. All models were trained on the same feature set (*fs_strong*) to ensure fair comparison. Evaluation was performed on both *Overall-all* (all days) and *Hotday-all* (extreme days) cohorts.

Note that for the TCN architecture, two variants were explored: (1) a *day-model* trained on daily 48-slot sequences, and (2) an *all-model* trained on continuous multi-day sequences to capture longer temporal dependencies. Since the *day-model* adopts the same 48-in, 48-out sequence format as the LSTM, it is used here for direct comparison across model families. The *all-model* is presented separately in **Appendix B**.

6.3.1 Overall Cohort Results

Table 8 presents the performance comparison of sequence-based (LSTM, TCN) and boosting-based (XGBoost, LightGBM) models trained on the same enriched feature set. LSTM with Heat-Aware Cost Weighting (HACW, $\alpha = 1.5$) and TCN, and XGBoost achieve comparable overall accuracy, all maintaining low error magnitudes (MAE $\approx$ 325–335 MW, RMSE $\leq$ 520 MW), confirming strong generalisation under normal operating conditions. In contrast, LightGBM shows markedly higher variance and instability, with RMSE exceeding 900 MW, indicating weaker calibration and reduced robustness under routine demand conditions.

Table 8. Aggregate performance of model families on the Overall-all cohort

Model	MAE (MW)	RMSE (MW)	MAPE (%)	Bias (MW)
LSTM (HACW $\alpha=1.5$)	335.00	477.20	4.29	66.62
TCN	323.83	509.31	4.01	-7.93
XGBoost	335.06	516.53	4.24	81.80
LightGBM	648.49	912.88	6.96	-320.66

6.3.2 Hot-Day Cohort Results

As shown in Table 9, evaluation of model robustness under heat-stress scenarios reveals distinct behaviours across model families. Sequence models (LSTM and TCN) and XGBoost maintain relatively stable accuracy despite the increased volatility of extreme-heat days. In contrast, LightGBM exhibits pronounced instability and over-sensitivity to temperature-driven demand spikes, resulting in large and erratic errors (RMSE $>$ 1100 MW).

Table 9. Aggregate performance of model families on the Hotday-all cohort

Model	MAE (MW)	RMSE (MW)	MAPE (%)	Bias (MW)
LSTM (HACW $\alpha=1.5$)	630.12	847.70	7.06	-250.37
TCN	613.91	890.97	6.54	-456.37
XGBoost	547.63	813.84	5.77	-332.38
LightGBM	1268	1107.60	13.85	27.21

Overall, sequence and boosting models deliver comparable short-term accuracy, but sequence architectures offer smoother performance and stronger resilience under extreme heat, while LightGBM remains the least stable.

6.3.3 Horizon-wise Error Behaviour Across Model Families

As shown in Figure 14, TCN’s curves are smoother, with low and stable MAE/RMSE at short horizons under normal conditions. On hot days, errors rise markedly in the same afternoon window (≈ 20 –35 slots). The bias trend is a long, steady slide into underestimation that reaches roughly $-1,200$ MW near slots 32~40. (Its all-model variant, which uses multi-day context, is discussed in Appendix B.)

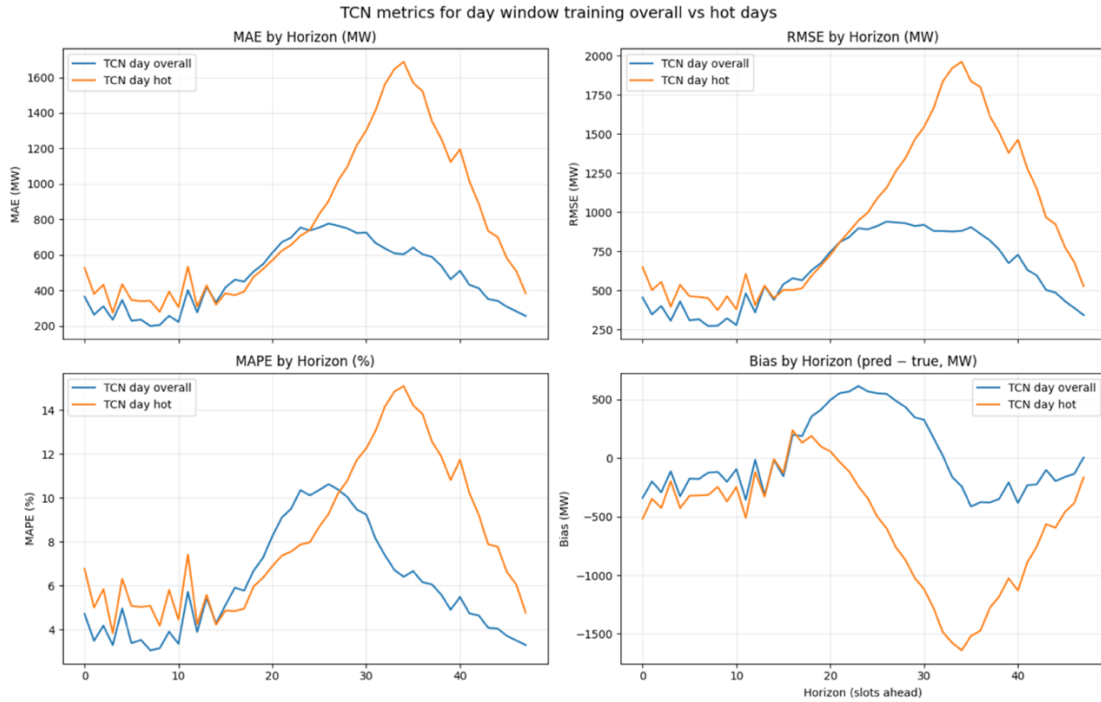


Figure 14. Horizon-wise MAE, RMSE, MAPE, and Bias curves for TCN under overall and hot-day cohorts

As demonstrated in Figure 15, while XGBoost maintains reasonable accuracy at short horizons, its error escalates sharply during mid-day (slot 24) to evening periods (slot 40). Both MAE and RMSE exceed 1200 MW at their peaks under hot-day conditions, indicating severe instability

when demand surges. The MAPE curve also rises steeply after 5am (slot 10), and the bias plot shows a strong downward deviation (underestimation) during the hottest periods. These results suggest that XGBoost's boosting-based structure, though efficient for normal day regimes, struggles to capture the additional demand spikes driven by heat stress.

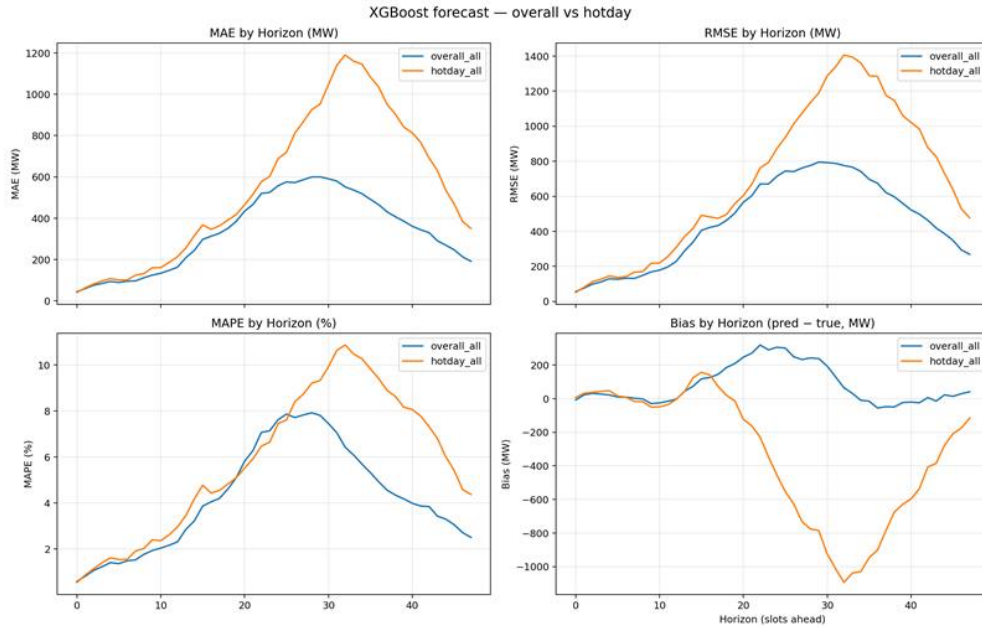


Figure 15. Horizon-wise MAE, RMSE, MAPE, and Bias curves for XGBoost under overall and hot-day cohorts

LightGBM — overall vs hotday (24-match)

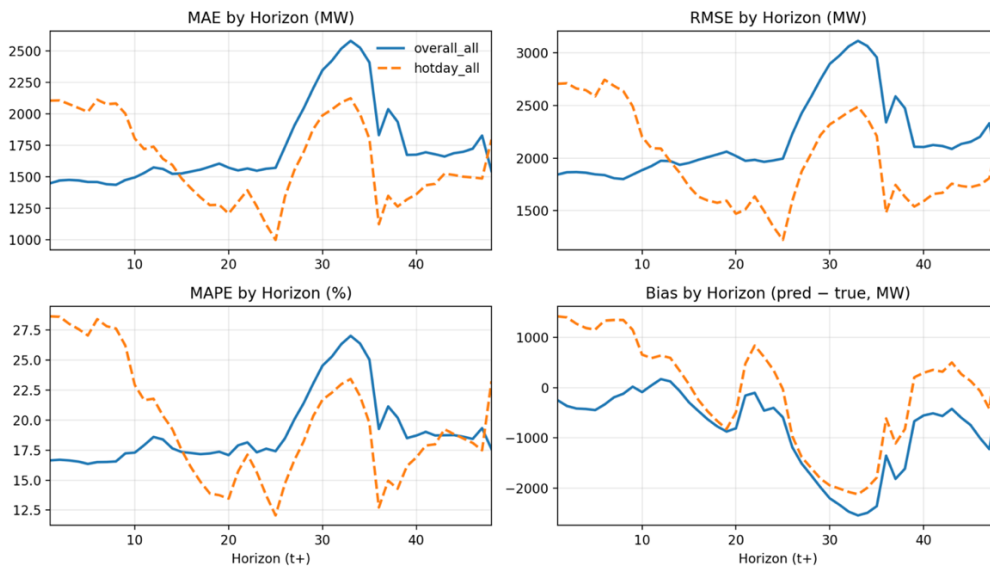


Figure 16. Horizon-wise MAE, RMSE, MAPE, and Bias curves for LightGBM under overall and hot-day cohorts

As shown in Figure 16, LightGBM exhibits irregular and unstable error behaviour across horizons. While short-horizon accuracy remains acceptable, performance deteriorates sharply beyond 20–25 slots, with large oscillations in both MAE and RMSE under hot-day conditions. The MAPE and bias curves further indicate inconsistent correction between over- and underestimation, highlighting LightGBM’s limited ability to adapt to nonlinear, temperature-driven demand spikes.

Conclusion (RQ3)

Across both cohorts, the results reveal clear differences between model families. Sequence models (LSTM and TCN) maintain stable, low-variance forecasts and smoother horizon-wise errors under both normal and heat-stress conditions. In contrast, boosting models (XGBoost and LightGBM) deteriorate sharply during afternoon peak-load periods, showing large error spikes and bias swings. Among them, the LSTM with Heat-Aware Cost Weighting ($\alpha = 1.5$) achieves the most balanced performance—preserving overall-day accuracy while markedly improving robustness under extreme heat.

7. Discussion

7.1 Feature Enrichment and Model Sensitivity

The results show that expanding the feature set substantially improves the LSTM’s forecasting accuracy and stability. Temporal lags and rolling statistics capture persistence and short-term variability in demand, while calendar and temperature features encode cyclical and environmental influence. Without these components, the minimal LSTM baseline (fs2_min) exhibits clear structural bias and weaker generalisation, indicating that feature-poor models fail to capture key temporal and weather-driven patterns in demand. These findings directly support **RQ1**, demonstrating that well-designed feature engineering enhances both short-term accuracy and the model’s robustness across varying temporal and weather-driven demand dynamics.

7.2 Heat-Aware Cost Weighting and Robustness

Our experiments show that **moderate reweighting** ($\alpha = 1.5$) improves model robustness during heatwaves without degrading baseline performance. By giving slightly higher weight to extreme-event samples helps the model anticipate large deviations while maintaining stability on normal days. In contrast, heavier weighting ($\alpha \geq 2.0$) leads to over-correction and loss of generalisation, highlighting the need for balance.

These results provide a practical insight: small, data-driven adjustments in the training objective can substantially strengthen reliability under stress conditions. From an operational viewpoint, this means grid forecasters can enhance resilience to heat-driven demand surges without redesigning their entire forecasting pipeline. This supports **RQ2**, showing that modest cost-weighting improves robustness with minimal trade-off.

7.3 Cross-Model Lessons

Addressing **RQ3**, the comparison across model families reveals distinct strengths and trade-offs. Boosting models such as XGBoost and LightGBM provide efficient short-term learning but degrade sharply under extreme conditions, showing higher variance and instability. In contrast, sequence-based models (LSTM and TCN) demonstrate stronger temporal awareness and stability, with the heat-weighted LSTM achieving the best balance between daily accuracy and resilience during heat stress. Overall, these results suggest that modest reweighting within sequence models offers a practical way to enhance operational robustness without increasing architectural complexity.

8. Conclusion and Future Work

This study demonstrates that enhancing feature design and applying moderate heat-aware weighting can make demand forecast more resilient to extreme heat without compromising day-to-day accuracy. Among all tested models, sequence-based approaches, particularly the LSTM with Heat-Aware Cost Weighting on hot-days ($\alpha = 1.5$), deliver the most reliable balance between accuracy, bias correction, and operational stability. These results suggest a practical solution for improving grid forecasting performance under increasing climate stress.

Future research may involve synthetic data generation for rare events, hybrid ensemble frameworks blending boosting and sequence models, and scalable evaluations across diverse regions to ensure operational resilience.

References

- Agbehadji, I.E. and Obagbuwa, I.C., (2024). *Systematic review of machine learning and deep learning techniques for spatiotemporal air quality prediction*. Atmosphere, 15(11), p.1352. <https://doi.org/10.3390/atmos15111352>
- Ajder, A., Hamza, H. and Ayaz, R. (2025). *Wavelet-Enhanced Hybrid LSTM–XGBoost model for predicting time series containing unpredictable events*. IEEE Access, pp. 1–11. <https://doi.org/10.1109/access.2025.3556540>
- Baharudin, Z.; Kamel, N. (2008) *Autoregressive Method in Short Term Load Forecast*. In Proceedings of the 2008 IEEE 2nd International Power and Energy Conference, Johor Bahru, Malaysia, pp. 1603–1608.
- Bai, S., Kolter, J. Z., Koltun, V. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling*. Available at: <https://doi.org/10.48550/arXiv.1803.01271>
- Bouktif, S., Fiaz, A., Ouni, A. and Serhani, M.A. (2018) *Optimal Deep Learning LSTM Model for Electric Load Forecasting using Feature Selection and Genetic Algorithm: Comparison with Machine Learning Approaches*. Energies, 11(7), 1636. Available at: <https://www.mdpi.com/1996-1073/11/7/1636>
- Dong, D., Wen, F., Zhang, Y. and Qiu, W. (2023). *Application of XGBoost in electricity consumption prediction*. 2023 IEEE 3rd International Conference on Electronic Technology, Communication and Information (ICETCI), pp. 1260–1264. <https://doi.org/10.1109/icetci57876.2023.10176934>
- Dudek, G. (2022). *A comprehensive study of Random Forest for short-term load forecasting*, Energies, vol. 15, no. 20, p. 7547.
- Elsayed, S., Thyssens, D., Rashed, A., Jomaa, H.S. and Schmidt-Thieme, L. (2021). *Do we really need deep learning models for time series forecasting?* arXiv preprint, arXiv:2101.02118. <https://arxiv.org/abs/2101.02118>
- Hamer, S., Sleeman, J., and Stajner, I. (2023) *Forecast-Aware Model Driven LSTM for Bias Correction in Air Quality Forecasting*. Environmental Modelling & Software.
- Hong, T. and Fan, S. (2016) ‘Probabilistic electric load forecasting: A tutorial review’, *International Journal of Forecasting*, 32(3), pp. 914–938. Available at: <https://doi.org/10.1016/j.ijforecast.2015.11.011>
- Jin, X., Dong, Y., Wu, J., Wang, J. (2010). *An Improved Combined Forecasting Method for Electric Power Load Based on Autoregressive Integrated Moving Average Model*. In Proceedings of the 2010 International Conference of Information Science and Management Engineering, Xi'an, China, Volume 2, pp. 476–480.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.Y., (2017). *LightGBM: A highly efficient gradient boosting decision tree*. *Advances in Neural Information Processing Systems*, 30, pp.3146–3154.

Khafaf, A., Tkiouat, M., El Ghmari, A. and Khalfi, A. (2019) *Application of Deep Learning Long Short-Term Memory in Energy Demand Forecasting*. In: *Proceedings of the 2019 7th International Renewable and Sustainable Energy Conference (IRSEC)*. IEEE, pp. 1–6. Available at: <http://arxiv.org/abs/1903.11941>

Lara-Benítez, P., Carranza-García, M., Luna-Romera, J. M., & Riquelme, J. C. (2020). *Temporal convolutional networks applied to energy-related time series forecasting*. *Applied Sciences*, 10(7), 2322.

Liu, H., Ma, Z., Yang, L., Zhou, T., Xia, R., Wang, Y., Wen, Q., and Sun, L. (2023) *SADI: A Self-adaptive Decomposed Interpretable Framework for Electric Load Forecasting under Extreme Events*. *Applied Energy*. Available at: <http://arxiv.org/abs/2306.08299>

Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2018). *The M4 Competition: Results, findings, conclusion and way forward*, *International Journal of Forecasting*, 34(4), pp. 802–808. Available at: <https://doi.org/10.1016/j.ijforecast.2018.06.001>

Malakouti, A., (2023). *LightGBM's effectiveness in rainfall prediction*. SSRN. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4397719 [Accessed 24 September 2025].

Mohamed, M., Mahmood, F.E., Abd, M.A., Rezkallah, M., Hamadi, A. and Chandra, A. (2023). *Load demand forecasting using eXtreme Gradient Boosting (XGBoost)*. 2023 IEEE Industry Applications Society Annual Meeting (IAS), pp. 1–7. <https://doi.org/10.1109/IAS54024.2023.10406613>

Rubattu, A., Massucco, S., Pitto, A., Silvestro, F. and Cossu, S., (2023). *A machine learning-based forecasting model for electrical load in presence of extreme temperature events*. *Applied Energy*, 329, p.120230.

Santágata, D.M., Castesana, P., Rössler, C.E. and Gómez, D.R. (2017). *Extreme temperature events affecting the electricity distribution system of the metropolitan area of Buenos Aires (1971–2013)*. *Energy Policy*, 106, pp. 404–414. Available at: <https://doi.org/10.1016/j.enpol.2017.04.006>

Sergio, A. and Colelli, F.P. (2025). *Weather-induced power plant outages: Empirical evidence from hydro and thermal generators in Europe*. *Energy Economics*, 148, 108549. Available at: <https://doi.org/10.1016/j.eneco.2025.108549>

Shabbir, N., Ahmadianhangar, R., Rosin, A., Jawad, M., Kilter, J. and Martins, J. (2023) *XGBoost based short-term electrical load forecasting considering trends & periodicity in historical data*, in *Proc. 2023 IEEE Int. Conf. on Energy Technologies for Future Grids (ETFG)*, Sydney, Australia, pp. 1–6.

Shaikh, A. K., Nazir, A., Khalique, N., Shah, A. S., & Adhikari, N. (2023). *A new approach to seasonal energy consumption forecasting using temporal convolutional networks*. Results in Engineering, 19, 101296.

Appendix

A. Evaluation Metrics and Aggregation Procedure

Model performance is assessed using four standard error metrics: **Mean Absolute Error (MAE)**, **Root Mean Square Error (RMSE)**, **Mean Absolute Percentage Error (MAPE)**, and **Bias** (prediction minus truth). These metrics are defined as follows, where $\hat{y}$ represents the predicted value and y represents the true value:

$$\begin{aligned} \text{MAE} &= \frac{1}{N} \sum_{t=1}^N |\hat{y}_t - y_t| & \text{MAPE} &= \frac{100}{N} \sum_{t=1}^N \left| \frac{\hat{y}_t - y_t}{y_t} \right| \\ \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{t=1}^N (\hat{y}_t - y_t)^2} & \text{Bias} &= \frac{1}{N} \sum_{t=1}^N (\hat{y}_t - y_t) \end{aligned}$$

Slot-wise vs Horizon-wise Evaluation

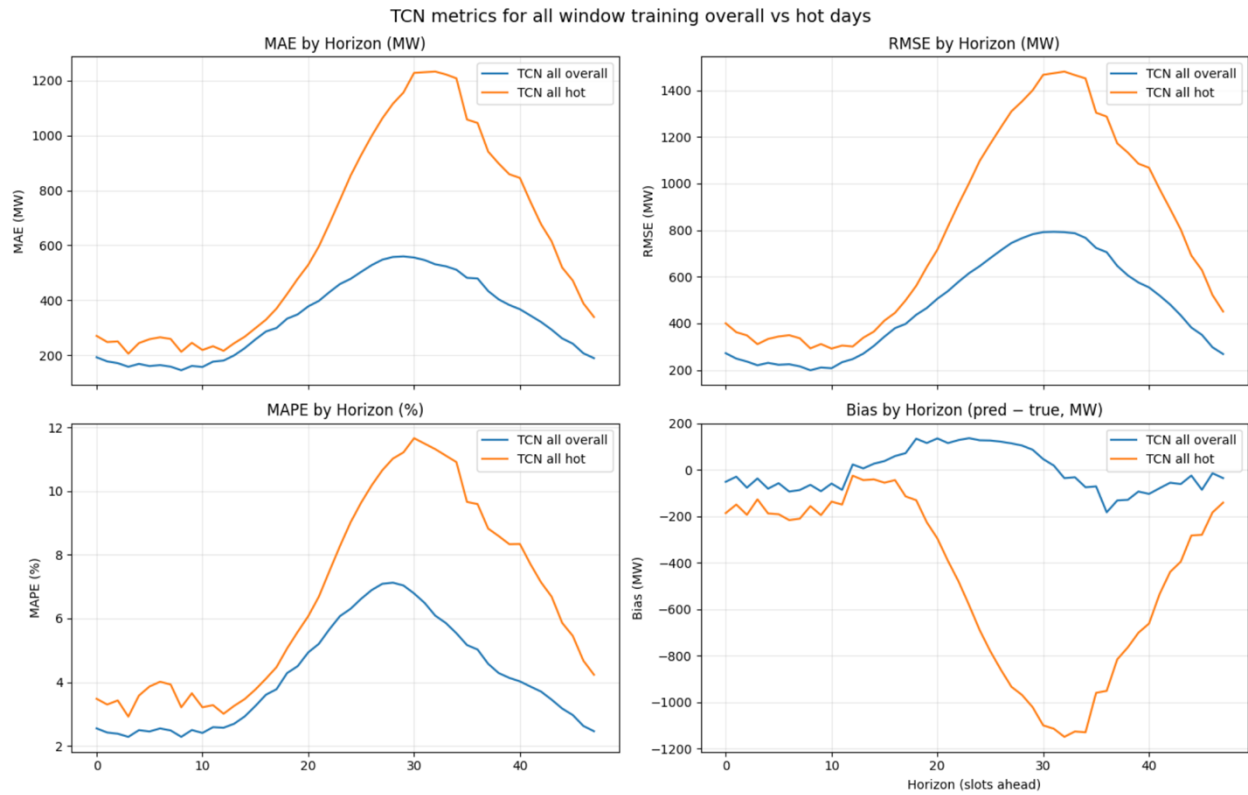
The evaluation procedure differs between operator forecasts and model forecasts. For **operator forecasts** (and other non-sequence models), a single prediction is issued directly for each half-hour slot of the day (48 slots). For example, for slot 1 (00:00–00:30), we aggregate all operator errors at this time across days and for slot 24 (11:30–12:00), we do the same, showing which times of day are hardest to predict. Since there is no concept of “lead time” or forecast horizon, metrics are evaluated **slot-wise**, aggregating errors for each time-of-day across the dataset. This highlights which hours of the day are inherently harder to predict and reveals systematic daily patterns of forecast difficulty.

In contrast, **sequence-based models** generate predictions at multiple horizons (1–48 half-hour steps ahead). Metrics are therefore evaluated **horizon-wise**, where errors are aggregated by lead time across all days. For example, at horizon 1, we aggregate all “1-step ahead” predictions across days and at horizon 24, we aggregate all “24-step ahead” forecasts, showing how accuracy falls as lead time grows. This provides insight into how forecast accuracy degrades as predictions are made further ahead in time.

Because the sequence models in this study are trained to output all 48 half-hour slots of the next day, their horizon-wise evaluations also produce 48 aggregated values per day, making the curves appear structurally similar to operator slot-wise results. This similarity arises only in the one-day-ahead setting: if forecasts were extended to multiple days (e.g., 3-day or 1-week horizons), horizon-wise evaluation would yield much longer error profiles (e.g., 144 or 336 horizons), while slot-wise evaluation would remain fixed at 48 daily slots. Thus, while the motivations differ—time-of-day performance versus lead-time performance—the structural overlap in our setting allows direct visual comparison of operator forecasts and sequence-model forecasts on equal footing.

B. TCN variant

TCN (all model) – trained on the full continuous sequence (no daily reset), allowing longer temporal dependencies.



C. Project code files

Files containing the code for the exploratory data analysis and modelling discussed in this report can be found on GitHub at <https://github.com/mae-li-unsw/project>.